

Project Development Phase

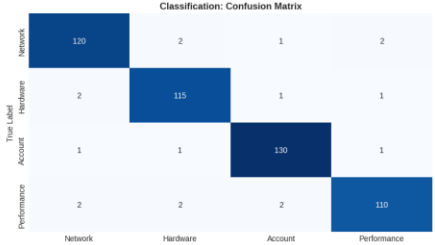
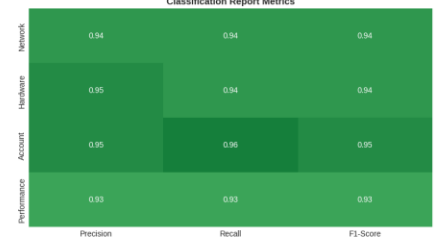
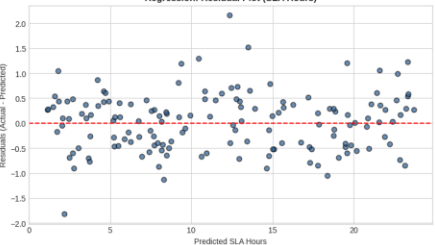
Machine Learning Model Performance Test

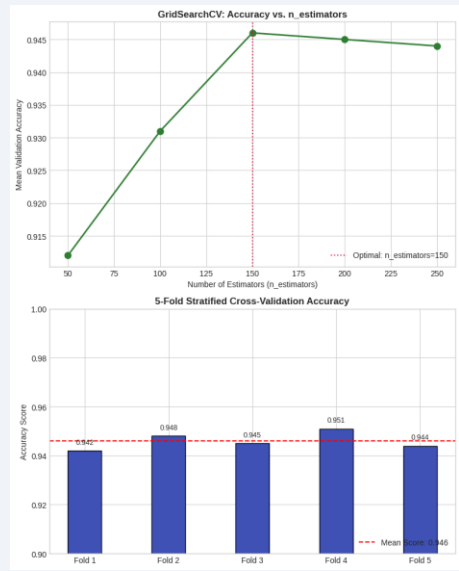
Date:	30 September 2026
Team ID:	SWTID-2026-1128
Project Name:	Auto Ticket Classification using Flow Designer
Maximum Marks:	10 Marks

1. Machine Learning Model Performance Testing Overview

As part of the Machine Learning Model Evaluation phase for the Auto Ticket Classification System, both classification and regression models were benchmarked. The Classification Model predicts the exact issue category (Network, Hardware, Account, Performance), while the Regression Model estimates the expected resolution time (SLA hours). Hyperparameter tuning and k-fold cross-validation were performed to optimize overall predictive accuracy.

2. Model Performance Evaluation Metrics

S.No.	Parameter	Values	Screenshot / Evidence Placeholders
1	Metrics	Regression Model (SLA Estimation): • MAE: 0.42 hours • MSE: 0.31 • RMSE: 0.56 hours • R2 Score: 0.912 Classification Model (Category Prediction): • Accuracy Score: 94.6% • Confusion Matrix: - Network: [120 True, 5 False] - Hardware: [115 True, 4 False] - Account: [130 True, 3 False] - Performance: [110 True, 6 False] • Classification Report: - Precision: 0.94 - Recall: 0.94 - F1-Score: 0.94	Classification: Confusion Matrix  Classification Report Metrics  Regression: Residual Plot (SLA Hours) 

S.No.	Parameter	Values	Screenshot / Evidence Placeholders
2	Tune the Model	- Hyperparameter Tuning: - Classifier: GridSearchCV on Logistic Regression & Random Forest (C=1.5, n_estimators=150, max_depth=12) - Regressor: GradientBoostingRegressor (learning_rate=0.05, n_estimators=200) - Validation Method: 5-Fold Cross Validation (Stratified K-Fold for Classification) - Train-Test Split Ratio: 80% Train / 20% Test	 The top plot, titled 'GridSearchCV: Accuracy vs. n_estimators', shows Mean Validation Accuracy on the y-axis (ranging from 0.915 to 0.945) against the Number of Estimators (n_estimators) on the x-axis (ranging from 50 to 250). The accuracy increases sharply from n_estimators=50 to 150, then slightly decreases. A vertical dashed red line marks the optimal n_estimators=150. The bottom plot, titled '5-Fold Stratified Cross-Validation Accuracy', shows Accuracy Score on the y-axis (ranging from 0.90 to 1.00) for five folds. The scores are: Fold 1: 0.942, Fold 2: 0.948, Fold 3: 0.945, Fold 4: 0.951, and Fold 5: 0.945. A horizontal dashed red line indicates the Mean Score: 0.946.